

JEE(Main + Advanced) : ENTHUSIAST COURSE : (SCORE-I)
ANSWER KEY
PAPER-1
PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8
	A.	B,C	A,B	A,D	A,C,D	A,B,C,D	A,C	A,C	A,B,C
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	4	9	2	6	4	5	2	3
SECTION-IV	Q.	1	2						
	A.	A→PRT,B→PQ,C→S,D→P	A→P,B→R,C→Q,D→R						

PART-2 : CHEMISTRY

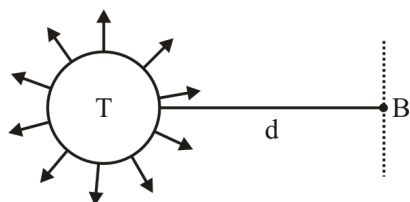
SECTION-I	Q.	1	2	3	4	5	6	7	8
	A.	B,C,D	A,B,C	A,C	A,C,D	A,B,C,D	A,B,C,D	B,C,D	A,C
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	16	5	3	5	415	19	10	6
SECTION-IV	Q.	1	2						
	A.	A→PR,B→RS,C→RS,D→QRT	A→RT,B→RT,C→QRS,D→QRS						

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8
	A.	A	B,C,D	A,C,D	C,D	A,C,D	A,B,C,D	A,B,C,D	B,D
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	4	2	54	8	4	2	6	100
SECTION-IV	Q.	1	2						
	A.	A→PR,B→Q,C→PQR,D→PQRST	A→S,B→P,C→PQRST,D→RST						

HINT – SHEET
PART-1 : PHYSICS
SECTION-I

1. Ans (B,C)


 Power radiates by sphere = $e\sigma AT^4$

 Power $\propto T^4$

intensity at point B =

$$\frac{\text{power radiated by sphere}}{4\pi d^2}$$

$$I \propto \frac{1}{d^2}$$

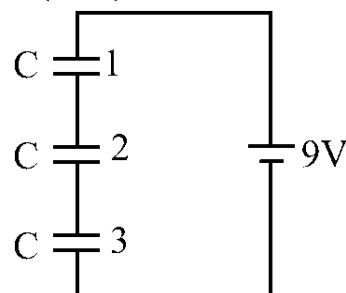
2. Ans (A,B)

 For points $r > R$, it will act as a point charge

$$\Rightarrow E \propto \frac{1}{r^2}$$

$$\text{for } r < R, E \propto \int f(r) dr \propto r^2$$

3. Ans (A,D)



Since, all capacitor are same,

$$\therefore V_2 = 3V$$

After K, is opened battery is out of circuit

 $\therefore$ no charge can flow through it.

4. Ans (A,C,D)

$$13.6z^2 \left[\frac{1}{1^2} - \frac{1}{(2n)^2} \right] = 204$$

$$13.6z^2 \left[\frac{1}{n^2} - \frac{1}{(2n)^2} \right] = 40.8$$

solving, $n = 2, z = 4$

5. Ans (A,B,C,D)

In standing wave energy is confined between node & antinode.

Node and antinode do not transfer its energy but element between node & antinode transfers its energy to other element.

6. Ans (A,C)

CB process is isothermal

$$3P_0V_1 = 2P_0V_0$$

$$V_1 = \frac{2V_0}{3}$$

BA process is adiabatic

$$PV^\gamma = \text{constant}$$

$$3P_0V_1^\gamma = P_0V_0^\gamma$$

$$\text{Workdone} = nRT \ln \left(\frac{V_2}{V_1} \right)$$

7. Ans (A,C)

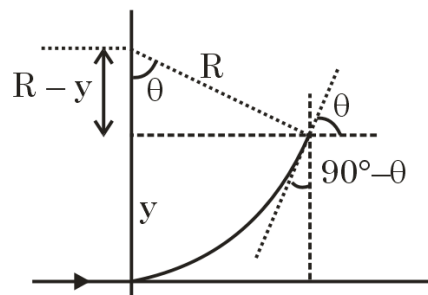
$$\text{Error in } \Delta T = \frac{1 \text{ sec}}{20} = 0.05 \text{ sec}$$

$$T = 2\pi\sqrt{\frac{\ell}{g}} \Rightarrow g \propto \frac{\ell}{T^2}$$

$$\Rightarrow \frac{\Delta g}{g} = \pm \left(\frac{2\Delta T}{T} + \frac{\Delta \ell}{\ell} \right)$$

$$\Rightarrow 100 \times \frac{\Delta g}{g} = \pm \left(\frac{2 \times 0.05}{2} + \frac{0}{1} \right) \times 100 = \pm 5\%$$

8. Ans (A,B,C)



$$\mu_0 \sin 90^\circ = \mu \sin (90^\circ - \theta)$$

$$\mu = \frac{\mu_0}{\cos \theta} = \frac{1}{\cos \theta} = \frac{R}{R - y}$$

$$\text{at } y = \frac{R}{2}; \mu = \frac{R}{R/2} = 2; \cos \theta = \frac{1}{2}$$

$$\theta = 60^\circ$$

$$\vec{R} = \hat{i} \cos 60^\circ + \hat{j} \sin 60^\circ$$

$$\vec{R} = \frac{\hat{i}}{2} + \frac{\sqrt{3}}{2} \hat{j}; \cos \theta = \frac{1}{\mu}$$

$$\cos \theta_{\max} = \frac{1}{2.5} = \frac{2}{5}$$

PART-1 : PHYSICS

SECTION-II

1. Ans (4)

$$T_1 = \frac{2(2m_1)m_1}{3m_1}g = \frac{4m_1}{3}g$$

$$T_2 = \frac{2(4m_2)(2m_2)}{6m_2}g = \frac{8}{3}m_2g$$

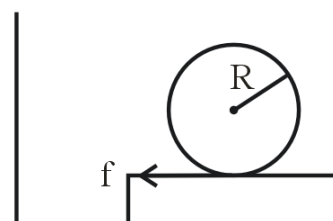
$$2T_1 \frac{\ell}{3} = 2T_2 \left(\frac{2\ell}{3} \right)$$

$$T_1 = 2T_2$$

$$\frac{4m_1}{3}g = 2 \times \frac{8}{3}m_2g$$

$$m_1 = 4m_2$$

2. Ans (9)



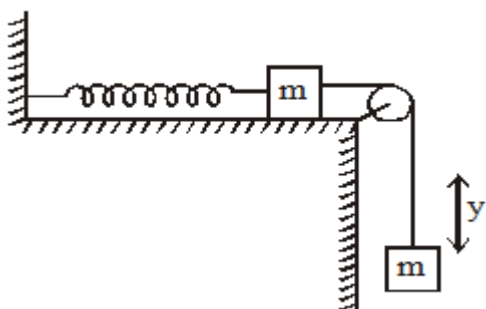
$$\alpha = \frac{2f}{mR}; a_{CM} = \frac{f}{m}$$

$$a_{POC} = \frac{3f}{m} = A(10)^2 \cos(10)t$$

for no slipping

$$100A = 3\mu g = 9 \Rightarrow A = 9 \text{ cm}$$

3. Ans (2)



$$mgy = \frac{1}{2}ky^2 + \mu mgy$$

$$\Rightarrow mg = \frac{1}{2}ky + \mu mg$$

4. Ans (6)

$$\text{Capillary height } h = \frac{2T \cos \theta}{rpg}$$

$$\Rightarrow h = 59.6 \text{ mm}$$

here 59.6 mm is greater than the protruding part hence water will rise in the capillary of insufficient height 25 mm.

$$\text{Now, } R = \frac{2T}{hpg} = 0.6 \text{ mm}$$

5. Ans (4)

$$i = 3 + 4 \sin\left(\omega t + \frac{\pi}{3}\right)$$

$$i^2 = 9 + 16 \sin^2\left(\omega t + \frac{\pi}{3}\right)$$

$$+ 24 \sin\left(\omega t + \frac{\pi}{3}\right) \langle i^2 \rangle$$

$$= 9 + 16 \left[\frac{1}{2} \right] + 24 [0]$$

$$= 9 + 16 \left(\frac{1}{2} \right) = 17$$

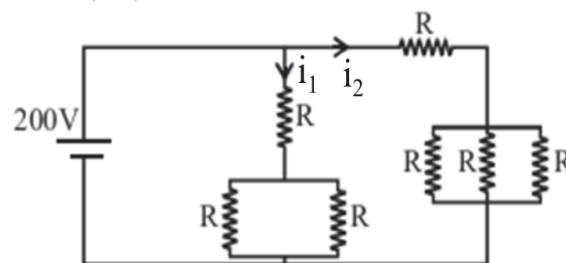
$$= \text{r.m.s. value} = \sqrt{17}$$

6. Ans (5)

In given circuit bulb D glow brightest

$$R = \frac{(200)^2}{100}$$

$$i_2 = \frac{200}{\left(\frac{4R}{3}\right)}; \text{Power in bulb D} = i_2^2 R$$



$$\Rightarrow P = \left(\frac{\frac{200}{\frac{4R}{3}}}{3}\right)^2 \times \frac{9}{16} \frac{(200)^2}{\frac{(200)^2}{100}} = \frac{900}{16} \text{ W}$$

$$\Rightarrow \frac{2}{3} \sqrt{P} = \frac{2}{3} \sqrt{\frac{900}{16}} = 5$$

7. Ans (2)

Consider a small area (shaded strip)

here E_{self} = Gravitational field due to this strip

and E_{ext} = Gravitational field due to the rest of spherical shell.

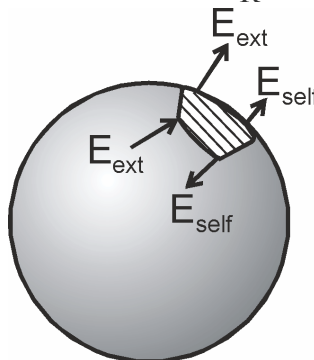
E_{in} = Gravitational field just inside the strip

E_{out} = Gravitational field just outside the strip

$$E_{\text{in}} = E_{\text{ext}} - E_{\text{self}} = 0$$

$$\Rightarrow E_{\text{ext}} = E_{\text{self}}$$

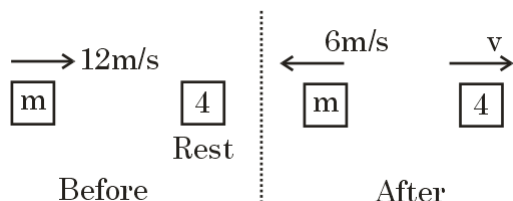
$$E_{\text{out}} = E_{\text{ext}} + E_{\text{self}} = \frac{GM}{R^2} \Rightarrow E_{\text{ext}} = \frac{GM}{2R^2}$$



After the shaded area has been removed there is no E_{self} and only E_{ext} .

$$\text{hence, } E_{\text{net}} = E_{\text{ext}} = \frac{GM}{2R^2}$$

8. Ans (3)



As the collision is elastic $e = 1$

$$12 + 0 = 6 + v$$

$$\Rightarrow v = 6 \text{ m/s}$$

Now, $V_{\text{COM (before)}} = V_{\text{COM (after)}}$

$$\Rightarrow \frac{12m}{m+4} = \frac{-6m+24}{m+4}$$

$$\Rightarrow m = \frac{4}{3} \text{ kg}$$

$$\therefore V_{\text{COM}} = \frac{12 \times \frac{4}{3}}{\frac{4}{3} + 4} = 3 \text{ m/s}$$

PART-1 : PHYSICS

SECTION-IV

2. Ans (A→P, B→R, C→Q, D→R)

$$\Delta x = S_2P - S_1P$$

S_2P increases so Δx increase

$$B = \frac{\Delta V}{d} \text{ does not change}$$

Central maxima will shift down word so net fringe pattern shift downward.

PART-2 : CHEMISTRY

SECTION-I

3. Ans (A, C)

The distance between nearest neighbours is same in both forms means '2r' is same in both and hence the unit cell edge length, i.e., cell constant, a , is different in both.

$$a_\alpha = \frac{4r}{\sqrt{3}} \text{ and } a_\beta = \frac{4r}{\sqrt{2}}$$

The distance between next nearest neighbours in both form is 'a', which is different.

Void space in BCC = 32%, FCC = 26%

In both forms, the required diameter of largest spheres = $(a - 2r)$ and 'a' is different

5. Ans (A, B, C, D)

Carbyl amine test is given by 1° amine both aliphatic and aromatic.

PART-2 : CHEMISTRY

SECTION-II

2. Ans (5)

Given : First order kinetics



$$t = 0 \quad p_0 = 800 \text{ of Hg}$$

$$t = 10 \quad 800 - P \quad \frac{P}{4} \quad \frac{P}{2}$$

$$800 - 600$$

$$= 200 \text{ mm of Hg}$$

total pr. after 10 min.

$$= 800 - P + \frac{P}{4} + \frac{P}{2} = 650$$

$$[P = 600] \text{ mm of Hg}$$

$$800 \xrightarrow{t_{1/2}} 400 \xrightarrow{t_{1/2}} 200$$

$$2t_{1/2} = 10 \text{ min}$$

$$[t_{1/2} = 5 \text{ min}]$$

6. Ans (19)

$$(a) = \text{KO}_2$$

$$(b) = \text{K}_2[\text{PtCl}_6]$$

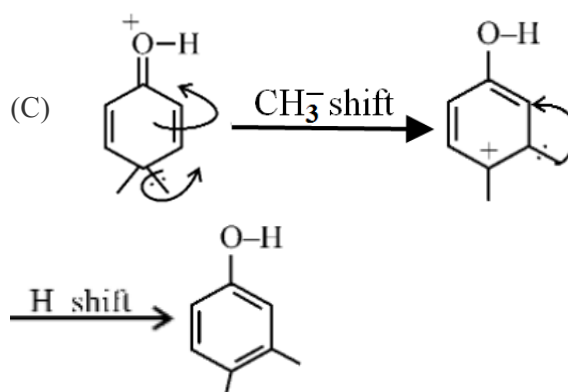
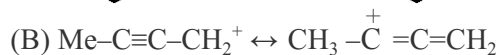
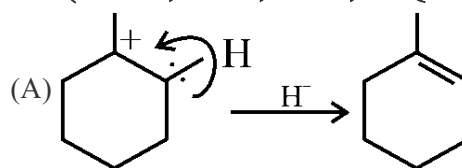
$$(c) = \text{MnO}_4^-$$

$$(d) = \text{O}_2$$

PART-2 : CHEMISTRY

SECTION-IV

1. Ans (A→PR, B→RS, C→RS, D→QRT)



PART-3 : MATHEMATICS

SECTION-I

1. **Ans (A)**

$$\begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} = \begin{vmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 3 & 1 & 2 \end{vmatrix} = 0$$

$$\Rightarrow (x-y)(y-z)(z-x) = 0$$

2. **Ans (B,C,D)**

The curve M is a circle of radius $\sqrt{2}$ unit with center at (1,1,1).

Point A is (2,0,1). And, line L is tangent to the circle M at point A.

3. **Ans (A,C,D)**

$$y' \sin x + y \cos x = x \sin x$$

$$\Rightarrow y' + y \cot x = x \Rightarrow y \sin x = \int x \sin x \, dx + c$$

$$\Rightarrow y \sin x = -x \cos x + \sin x + c$$

$$\Rightarrow y = 1 - x \cot x \quad \left(\text{as } y \left(\frac{\pi}{2} \right) = 1 \right)$$

$$\text{Now, } y'(x) = x \operatorname{cosec}^2 x - \cot x$$

$$= \frac{x - \sin x \cos x}{\sin^2 x} = \frac{2x - \sin 2x}{2 \sin^2 x} > 0 \quad \forall x \in \left(0, \frac{\pi}{2} \right]$$

$$\text{Hence, } y(x) \text{ is strictly increasing in } \left(0, \frac{\pi}{2} \right]$$

4. **Ans (C,D)**

$$(A) \text{ Since } \lim_{h \rightarrow 0} \frac{h \ln |h|}{h} \text{ does not exist.}$$

So, $f(x)$ is not differentiable at $x = 0$

$$(B) g(x) = \begin{cases} x^2 + 1; & x \in [0, 1) \\ x(x+1) + 1; & x \in (-1, 0) \end{cases}$$

Hence, $g(x)$ is not differentiable at $x = 0$

$$(C) f(g(x)) = (x\{x\} + 1) \ln(x\{x\} + 1); \text{ as } g(x) > 0$$

$$= \begin{cases} (x^2 + 1) \ln(x^2 + 1); & x \in [0, 1) \\ (x^2 + x + 1) \ln(x^2 + x + 1); & x \in (-1, 0) \end{cases}$$

At $x = 0$

$$\text{RHD} = 0, \text{ LHD} = 1$$

$$(D) f(g(x)) = \begin{cases} (x^2 + 1) \ln(x^2 + 1); & x \in [0, 1) \\ (x^2 - x + 1) \ln(x^2 - x + 1); & x \in [1, 2) \end{cases}$$

At $x = 1$

$$\text{RHD} = 1 \neq \text{LHD}$$

5. **Ans (A,C,D)**

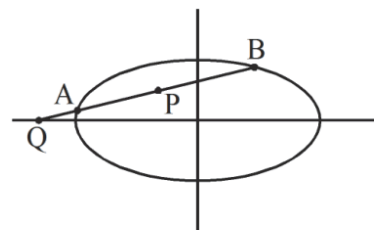
$$\tan^{-1}(\sqrt{2}-1) < \tan^{-1}(\sqrt{3}-1) < \tan^{-1} 1$$

$$\Rightarrow 4 \times \frac{\pi}{8} < \alpha < 4 \times \frac{\pi}{4} \Rightarrow \frac{\pi}{2} < \alpha < \pi$$

$$\text{Also, } \tan^{-1}(\sqrt{2}-1) < \tan^{-1} \frac{1}{2} < \tan^{-1} \frac{1}{\sqrt{3}}$$

$$\Rightarrow 8 \times \frac{\pi}{8} < \beta < 8 \times \frac{\pi}{6} \Rightarrow \pi < \beta < \frac{4\pi}{3}$$

6. **Ans (A,B,C,D)**



Let QA, QP and QB be in A.P.

$$\Rightarrow 2QP = QA + QB$$

$$\Rightarrow 2QP - 2QA = QB - QA$$

$$\Rightarrow 2AP = AB$$

$\Rightarrow P$ is the mid-point of chord AB

$\Rightarrow$ curve C_1 and C_2 are identical

Also, Let $P(h,k)$

$$\text{So, } xh + 2yk = h^2 + 2k^2$$

$$\Rightarrow -2h = h^2 + 2k^2$$

$$\Rightarrow x^2 + 2x + 2y^2 = 0$$

8. **Ans (B,D)**

$$f(x) = (e^{\sin x} + e^{-\sin x}) \cos x > 0; \forall x \in \left[0, \frac{\pi}{2} \right]$$

$$\Rightarrow f(x) \in \left[0, e - \frac{1}{e} \right]$$

$$\text{Also, } g(x) = \frac{\pi}{2}(1 - \cos x) \in \left[0, \frac{\pi}{2} \right]$$

$$\Rightarrow f(g(x)) \in \left[0, e - \frac{1}{e} \right]$$

$$\begin{aligned} \text{Now, } \lim_{x \rightarrow 0^+} \frac{f(x)}{f(g)} &= \lim_{x \rightarrow 0^+} \frac{e^{2 \sin x} - 1}{e^{\sin x} \frac{\pi}{2}(1 - \cos x)} \\ &= \frac{2}{\pi} \lim_{x \rightarrow 0^+} \frac{e^{2 \sin x} - 1}{2 \sin x} \times \frac{2 \sin x}{x^2} \times \frac{x^2}{1 - \cos x} = \infty \end{aligned}$$

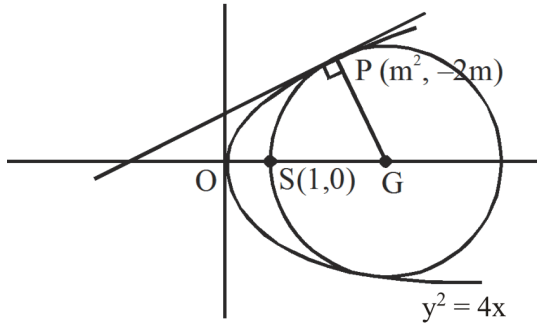
$$\text{Further, range of } g(f(x)) \text{ is } \left[0, \frac{\pi}{2} \left(1 - \cos \left(e - \frac{1}{e} \right) \right) \right]$$

$$\text{and } 0 < \frac{\pi}{2} < \frac{\pi}{2} \left(1 - \cos \left(e - \frac{1}{e} \right) \right)$$

PART-3 : MATHEMATICS

SECTION-II

1. **Ans (4)**



The required radius remains same
if we reflect our system about the line $y = x$.

$\overleftrightarrow{PG}$:-

$$y = mx - 2m - m^3$$

$$\Rightarrow G(m^2 + 2, 0)$$

$$\Rightarrow r = SG = m^2 + 1$$

$$\text{Also, } PG = SG$$

$$\Rightarrow 4 + 4m^2 = (m^2 + 1)^2$$

$$\Rightarrow m^2 + 1 = 4 = r$$

2. **Ans (2)**

Probability of success in a single round

$$= 1 - \left(\left(\frac{1}{2} \right)^7 + {}^7C_1 \left(\frac{1}{2} \right)^6 \times \frac{1}{2} \right)$$

$$= 1 - \frac{1}{16} = \frac{15}{16}$$

$$\text{Hence, } 1 - \left(\frac{1}{16} \right)^n \geq \frac{255}{256} \Rightarrow \frac{1}{16^2} \geq \frac{1}{16^n}$$

$$\Rightarrow n \geq 2$$

3. **Ans (54)**

$$\begin{aligned} & \int_{-3}^0 g([x]) \, dx + \int_0^2 g(x) \, dx \\ &= \int_{-3}^0 ([x] + 1)([x] + 2)([x] + 3) \, dx + \int_0^2 (x + 1)(x + 2)(x + 3) \, dx \\ &= 0 + \int_{-1}^1 (t + 2)(t + 3)(t + 4) \, dt \quad (\text{put } x - 1 = t) \\ &= \int_{-1}^1 (9t^2 + 24) \, dt = 2(3 + 24) = 54 \end{aligned}$$

4. **Ans (8)**

Let box dimension be $\ell \times \ell \times h$.

$$\text{So, } \ell^2 h = 8^3$$

$$V = (\ell + 4)^2 (h + 4) - \ell^2 h - (2\ell^2 + 4\ell h) \times \frac{1}{2} \times 2$$

$$= 2 \left(\ell^2 + 16\ell + \frac{2 \times 8^3}{\ell} + \frac{8^4}{\ell^2} + 32 \right)$$

$$= 2 \left(\ell^2 + 8\ell + 8\ell + \frac{8^3}{\ell} + \frac{8^3}{\ell} + \frac{8^4}{\ell^2} + 32 \right) \geq$$

$$2 \times (6 \times 8^2 + 32)$$

$$(\because \text{A.M.} \geq \text{G.M.})$$

Further, equality holds when $\ell = 8$

$$\text{i.e. } h = 8$$

5. **Ans (4)**

$$4\sin^4 x + 4\sin^3 x + 4\sin^2 x = \sin^2 x + \sin x + 1$$

$$\Rightarrow 4\sin^2 x (\sin^2 x + \sin x + 1) = \sin^2 x + \sin x + 1$$

$$\Rightarrow \sin x = \pm \frac{1}{2}$$

6. **Ans (2)**

$$\begin{aligned} & \frac{64}{\pi^2} \int_0^1 \frac{\tan^{-1} \left(\frac{2x}{x+1} \right)}{(2x)^2 + (x+1)^2} \, dx \\ &= \frac{64}{\pi^2} \int_0^1 \frac{\tan^{-1} \left(\frac{2x}{x+1} \right)}{1 + \left(\frac{2x}{x+1} \right)^2} \frac{dx}{(x+1)^2} \\ &= \frac{32}{\pi^2} \left[\left(\tan^{-1} \left(\frac{2x}{x+1} \right) \right)^2 \right]_0^1 \\ &= \frac{32}{\pi^2} \left(\frac{\pi^2}{16} - 0 \right) = 2 \end{aligned}$$

7. **Ans (6)**

$$\begin{aligned} \sum_{1 \leq i < j \leq 6} |\alpha_j - \alpha_i| &= 6|\alpha_2 - \alpha_1| + 6|\alpha_3 - \alpha_1| + 3|\alpha_4 - \alpha_1| \\ &= 6 \times 1 + 6 \times \sqrt{3} + 3 \times 2 = 6(2 + \sqrt{3}) \end{aligned}$$

8. **Ans (100)**

$$x^2 \rightarrow (1 + 2x + x^2)^{50} (1 - x)^{100}$$

$$\Rightarrow x^2 \rightarrow (x + 1)^{100} (1 - x)^{100} = (1 - x^2)^{100}$$

$$\Rightarrow k = -100$$

PART-3 : MATHEMATICS

SECTION-IV

1. Ans (A->PR,B->Q,C->PQR,D->PQRST)

(A) Required $\hat{r}$ is along

$$\frac{\hat{i} + \hat{j}}{\sqrt{2}} + \frac{-\hat{i} + \hat{j}}{\sqrt{2}} + \hat{k}$$

(as these three vectors are mutually perpendicular unit vectors)

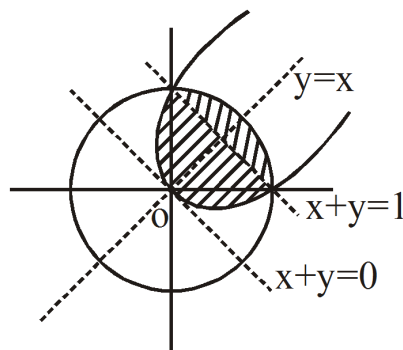
$$\text{So, } \hat{r} = \pm \frac{\sqrt{2}\hat{j} + \hat{k}}{\sqrt{3}}$$

$$(B) f(x) = |x| - |x| - 1 = -1$$

$$(C) S = \sum_{r=1}^{10} (r+1)(r^2-r+1) = \sum_{r=1}^{10} (r^3+1)$$

$$(D) S_n = \sum_{r=1}^n \left(\frac{n^2-n}{2} + r \right)^3$$

2. Ans (A->S,B->P,C->PQRST,D->RST)



$$(A) \text{ Let } b \sin C + c \sin B = \alpha \quad \dots (i)$$

$$\text{we have } b \cos C + c \cos B = a \quad \dots (ii)$$

$$(i)^2 + (ii)^2 :-$$

$$b^2 + c^2 + 2bc \cos(B-C) = a^2 + a^2$$

$$\Rightarrow a^2 + a^2 \leq (b+c)^2$$

$$\Rightarrow a^2 \leq (b+c)^2 - a^2 = (b+c+a)(b+c-a)$$

$$\Rightarrow a^2 \leq (2S) 2(S-a)$$

$$\Rightarrow a^2 \leq 4 \frac{\Delta}{\tan(A/2)} = \frac{4\sqrt{3}}{\sqrt{3}} = 4$$

$$\Rightarrow a \leq 2 \text{ (equality holds when } B=C)$$

$$(B) \tan\left(\frac{B-C}{2}\right) = \frac{b-c}{b+c} \cot \frac{A}{2}$$

$$\Rightarrow \tan\left(\frac{B-C}{2}\right) \tan \frac{A}{2} = \frac{1}{2}$$

$$(C) |[\vec{a} \ \vec{b} \ \vec{c}]| = |2 \sin 2\theta + \cos 2\theta - 1| \in [0, \sqrt{5}+1]$$

$$(D) \left(\frac{x-y}{\sqrt{2}}\right)^2 = \sqrt{2} \left(\frac{x+y}{\sqrt{2}}\right)$$

Require Area

$$= \frac{2}{3} \left(\sqrt{2} \times \frac{1}{\sqrt{2}} \right) + \frac{\pi}{4} - \frac{1}{2} \times 1 \times 1 = \frac{\pi}{4} + \frac{1}{6}$$

JEE(Main + Advanced) : ENTHUSIAST COURSE : (SCORE-I)
ANSWER KEY
PAPER-2
PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	B	B	B	A	C	C	A	C	B
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	D	D	A	B	C	D	C	B	A	A
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	4.00	16.00	4.80	10.00	0.30	42.00	10.00	6.00	35.00	2.00

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	D	D	B	C	C	D	C	B	B
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	B	C	D	B	A	C	D	D	B	C
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	6.00	5.00	3.00	5.00	3.00	4.00	1.33	20.00	62.50	7.50

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	C	D	C	D	B	A	B	B	C
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	A	C	C	B	D	D	A	C	D	A
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	9.50	23.33	5.00	3.99	1.50	8.50	3.75	0.00	4.00	4.00

HINT – SHEET
PART-1 : PHYSICS
SECTION-I

 1. **Ans (C)**

$$E = h\nu = 54.4 \text{ eV}$$

 Ionization energy of given atom = $4 \times 13.6 = 54.4 \text{ eV}$

Hence electron will become free and no spectral lines will be observed.

 2. **Ans (B)**

Mass of radius

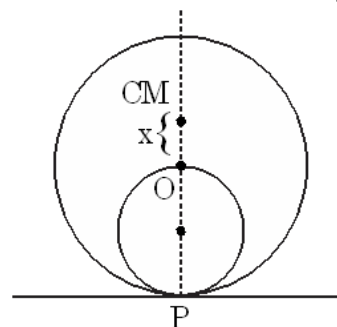
$$\frac{R}{2} = \frac{M}{\left(\frac{4\pi}{3}R^3 - \frac{4\pi}{3} \times \frac{R^3}{8}\right)} \times \frac{4\pi}{3} \frac{R^3}{8} = \frac{M}{7}$$

$$\text{Mass of complete sphere } M + \frac{M}{7} = \frac{8M}{7}$$

Distance of COM from O = x

$$x \cdot M = \frac{M}{7} \cdot \frac{R}{2} \therefore x = \frac{R}{14}$$

The sphere may be assumed to be in pure rotation about an axis through contact point P



$$V_{CM} = \omega \cdot (R + x) = \frac{V}{R} \left(R + \frac{R}{14} \right) = \frac{15V}{14}$$

 3. **Ans (B)**

 Beat frequency Δf must be less than 1 in 10s.

$$\Rightarrow \Delta f < 0.1 \text{ Hz}$$

 $\therefore$ Difference in frequencies of the two sources is less than 0.1 Hz.

$$\therefore 1023.9 \text{ Hz} < f_{\text{string}} < 1024.1 \text{ Hz}$$

4. **Ans (B)**

$$I_B R_B + V_{BE} = 2V$$

As V_{BE} is negligible

$$\therefore I_B = \frac{2V}{R_B} = \frac{2V}{100 \text{ k}\Omega} = 20 \mu\text{A}$$

Collector current, $I_C = \beta I_B = 50 \times 20 \mu\text{A} = 1000 \mu\text{A} = 1\text{mA}$

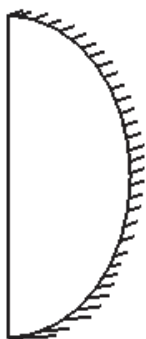
Applying Kirchhoff's voltage law to the circuit DEN, we get,

$$I_C R_C + V_{CE} = 9$$

$$\text{or } 1 \text{ mA} + 2\text{k}\Omega + V_{CE} = 9$$

$$\text{or } V_{CE} = 9 - 2 = 7V$$

5. **Ans (A)**



$$P_{eq} = 2P_L + P_m$$

$$\Rightarrow P_{eq} = 2 \left(\frac{3}{2} - 1 \right) \left(\frac{1}{\infty} - \frac{1}{-30} \right) - \frac{1}{-30/2}$$

$$\Rightarrow P_{eq} = \frac{1}{30} + \frac{2}{30} = \frac{1}{10} \Rightarrow -\frac{1}{f_{eq}} = \frac{1}{10} \Rightarrow f_{eq} = -10$$

i.e., it will behave as concave mirror of focal length 10cm. To form real image of same size, object should be placed at center of curvature.

6. **Ans (C)**

Let x, y and z direction be as shown.

$$V_A - V_0 = -E_x (1\text{m})$$

$$E_x = 1 \text{ V/m}$$

$$V_B - V_0 = -E_x(3\text{m}) - E_y (2\text{m}) - E_z (1\text{m})$$

$$-6 = -3E_x - 2E_y - E_z$$

$$2E_y + E_z = 3$$

$$\text{And } V_C - V_0 = -E_x (1\text{m}) - E_y (3\text{m}) - E_z (4\text{m})$$

$$-3 = -1 - 3E_y - 4E_z \Rightarrow 3E_y + 4E_z = 2$$

Solving (2) and (3)

$$E_y = 2 \text{ V/m}$$

$$E_z = -1 \text{ V/m}$$

7. **Ans (C)**

$$T_1 = 2\pi \sqrt{\frac{I_1 + I_2}{M_1 + M_2}} \quad (i)$$

$$\text{and } T_2 = 2\pi \sqrt{\frac{I_1 + I_2}{M_1 - M_2}} \quad (ii)$$

Using (i) and (ii)

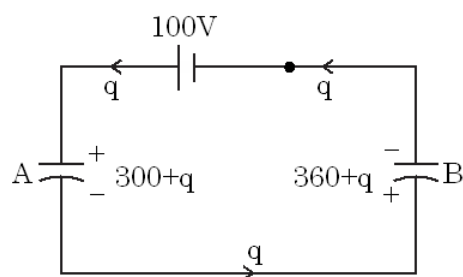
$$\frac{M_1}{M_2} = \frac{T_2^2 + T_1^2}{T_2^2 - T_1^2} = \frac{4^2 + 3^2}{4^2 - 3^2} = \frac{25}{7}$$

8. **Ans (A)**

Charges on two capacitors are

$$q_A = 100 \times 3 = 300 \mu\text{C}$$

$$\text{and } q_B = 180 \times 2 = 360 \mu\text{C}$$



After the switch S is closed let charge q flow in the circuit as shown in the figure.

Using Kirchhoff's voltage law we can write

$$\frac{360 + q}{2} + \frac{300 + q}{3} = 100$$

$$\Rightarrow 360 \times 3 + 3q + 600 + 2q = 600$$

$$\Rightarrow q = -216 \mu\text{C}$$

$\therefore$ Final charge on the two capacitors is

$$q'_A = 300 + q = 84 \mu\text{C}$$

$$q'_B = 360 + q = 144 \mu\text{C}$$

9. **Ans (C)**

$$W_B = U_{\text{capacitor}} + U_{\text{surface energy}}$$

$$\Rightarrow Q \cdot V = \frac{1}{2} CV^2 + 2\sigma A$$

$$\Rightarrow \frac{CV^2}{2} = 2\sigma A$$

$$\Rightarrow V^2 = \frac{4\sigma A}{\epsilon_0 \epsilon_r} \times d = \frac{4\sigma d}{\epsilon_0 \epsilon_r}$$

$$\therefore V = 2 \sqrt{\frac{\sigma d}{\epsilon_0 \epsilon_r}}$$

10. **Ans (B)**

$$I_A \cos^2 \theta = I_B \cos^2 (90 - \theta)$$

$$\Rightarrow \tan^2 \theta = \frac{I_A}{I_B} = \frac{1}{3}$$

$$\therefore \theta = 30^\circ$$

11. **Ans (D)**

Distribute current using series parallel method.

12. **Ans (D)**

$$Q = Q_0 \cos \omega t$$

$$i = Q_0 \omega \sin \omega t$$

$$\omega = \frac{1}{\sqrt{LC}}$$

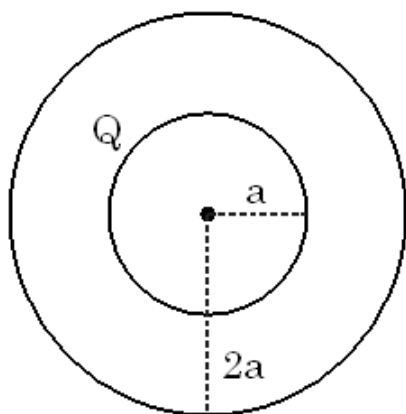
13. **Ans (A)**

$$W = Mg(h_1 + h_2 + h_3) + Mg(\mu_1 \ell_1 + \mu_2 \ell_2 + \mu_3 \ell_3)$$

14. **Ans (B)**

$$H_{\max} = \frac{V^2}{2g}$$

15. **Ans (C)**

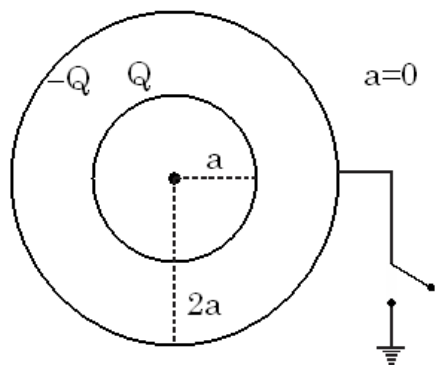


$$U_1 = \frac{Q^2}{2C}$$

$$C = 4\pi\epsilon_0 a$$

$$\Delta H = U_1 - U_2$$

$$\Delta H = \frac{1}{4\pi\epsilon_0} \cdot \frac{Q^2}{4a}$$



$$U_2 = \frac{Q^2}{2C'}$$

$$C' = \frac{4\pi\epsilon_0 (2a) \times a}{(2a - a)}$$

16. **Ans (D)**

$$\vec{\mu} = I \vec{A}$$

17. **Ans (C)**

Hint (QF) (B.W) = constant

18. **Ans (B)**

$$I = \frac{1}{2} \epsilon_0 E_0^2 C$$

19. **Ans (A)**

$$L.C. = \frac{0.5}{100} \text{ mm}$$

$$\frac{\Delta Y}{Y} = \frac{\Delta \ell}{\ell} + 2 \frac{\Delta d}{d}$$

$$\frac{\Delta \ell}{\ell} = \frac{0.005}{0.25} = \frac{1}{50}$$

$$2 \frac{\Delta d}{d} = \frac{2 \times 0.005}{0.5} = \frac{1}{50}$$

20. **Ans (A)**

$$23T = Kx$$

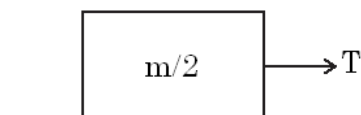
$$K_{eq} = \frac{K}{23}$$

PART-1 : PHYSICS

SECTION-II

1. **Ans (4.00)**

F.B.D. of first half of rod is



$$\because m = \rho AL$$

$$\therefore T = \frac{m}{2} \cdot a$$

$$T = \frac{\rho AL}{2} \cdot a$$

$$\Rightarrow \frac{T}{A} = \frac{\rho \ell a}{2}$$

$$\Rightarrow \text{Stress} = 10,000$$

2. **Ans (16.00)**

$$\because i = neAv_d$$

$$\Rightarrow v_d = \frac{i}{neA} \dots (1)$$

$$\because n' = \frac{n}{4}$$

$$A' = \frac{A}{4} \left\{ \because r' = \frac{r}{2} \right\}$$

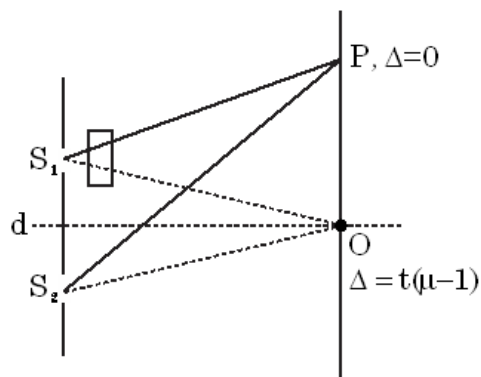
$$\therefore v'_d = \frac{i}{\frac{n}{4} \cdot e \cdot \frac{A}{4}} = 16 \cdot \frac{i}{neA} = 16v_d$$

3. **Ans (4.80)**

$$\frac{1}{t} = \frac{1}{t_1} + \frac{1}{t_2}$$

$$\text{or } t = \frac{t_1 t_2}{t_1 + t_2} = \frac{24 \times 6}{24 + 6} = 4.8 \text{ h}$$

4. **Ans (10.00)**



At 'O'

$$B = t(\mu - 1)$$

At point P

$$\Delta = \frac{dx}{D} - t(\mu - 1) = 0$$

$$\Rightarrow \frac{dx}{D} = t(\mu - 1)$$

$$\Rightarrow x = \frac{D}{d} t(\mu - 1)$$

$$= \frac{0.5 \times 1.5 \times 10^{-3}}{3 \times 10^{-3}} \times (1.4 - 1)$$

$$= 0.1 \text{ m}$$

$$= 10 \text{ cm}$$

5. **Ans (0.30)**

Let ϵ is the emissivity of the sphere then by Stefan's law.

$$E = \epsilon \sigma A (T^4 - T_0^4)$$

$$\text{or } 210 = \epsilon \sigma A (500^4 - 300^4) \dots\dots\dots(i)$$

When sphere is blackened, it behaves like a perfectly blackbody, so we have-

$$700 = \sigma A (500^4 - 300^4) \dots\dots\dots(ii)$$

Dividing equation (i) and (ii), we get

$$\epsilon = 0.3$$

6. **Ans (42.00)**

$$Q = nC_p \times 30 = n \times \frac{7R}{2} \times 30$$

$$\text{and } Q = n \times \frac{5R}{2} \times \Delta T$$

$$\therefore n \times \frac{7R}{2} \times 30 = n \times \frac{5R}{2} \times \Delta T$$

$$\text{or } \Delta T = 42 \text{ K}$$

7. **Ans (10.00)**

Tolerance value for silver = 10%

8. **Ans (6.00)**

$$a \sin \theta = \lambda$$

$$\Rightarrow a \times \sin 30^\circ = \lambda$$

$$\Rightarrow a = 2\lambda$$

$$a = 6\text{\AA}$$

9. **Ans (35.00)**

The number of nuclei present is given by $N = 0.05N_0$

$$N = N_0 e^{-\lambda t} \rightarrow t = -\frac{\ln N/N_0}{\lambda}$$

$$= -\frac{T_{1/2} \ln N/N_0}{0.693} = -\frac{(8.0d) \ln 0.05}{0.693} = 34.58d = \boxed{35d}$$

10. **Ans (2.00)**

$$\frac{V_{\text{output}}}{V_{\text{input}}} = \frac{1}{5}$$

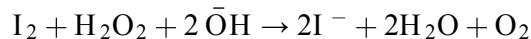
$$\Rightarrow V_{\text{output}} = \frac{100}{5} = 20 \text{ volt}$$

$$\therefore I = \frac{20}{10} = 2A$$

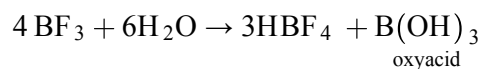
PART-2 : CHEMISTRY

SECTION-I

1. **Ans (D)**



5. **Ans (C)**



7. **Ans (D)**

Switzer's reagent $[Cu(NH_3)_4]SO_4$

IUPAC name - tetraammine copper (II) sulphate

8. **Ans (C)**

$$n_{\text{Na}_2\text{CO}_3} = M \times v = 0.25 \times 0.5 = 0.125$$

$$n_{\text{H}_2\text{CO}_3} = M \times v = 0.25 \times 0.5 = 0.125$$



$$t = 0 \quad 0.125 \text{ mol} \quad 0.125 \text{ mol}$$

$$t = t \quad 0 \quad 0 \quad 0.25 \text{ mol}$$

NaHCO_3 is Amphoteric salt.

$$\text{pH} = \frac{1}{2}(\text{p}K_{a_1} + \text{p}K_{a_2})$$

$$= \frac{1}{2}(7 + 11)$$

$$= 9$$

9. **Ans (B)**

$$\Delta S^\circ = S_{\text{CH}_4}^\circ - 2S_{\text{H}_2}^\circ - S_{\text{C}}^\circ$$

$$= 186.2 - 2 \times 130.6 - 6$$

$$= -81 \text{ J/K}$$

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ = -T\Delta S_{\text{uni}}$$

$$-75000 - 300(-81) = -300 \Delta S_{\text{uni}}$$

$$\therefore \Delta S_{\text{uni}} = 169$$

10. **Ans (B)**

$$\frac{r^+}{r^-} = \frac{126}{216} = 0.58$$

Ag^+ will occupy octahedral void.

$$\text{So } r^+ + r^- = \frac{a}{2}$$

$$126 + 216 = \frac{a}{2}$$

$$\therefore a = 684 \text{ pm}$$

11. **Ans (B)**

$$\lambda_m = \frac{1000 \times K}{M}$$

$$= \frac{1000 \times \frac{1}{R} \frac{\ell}{A}}{M}$$

$$= \frac{1000 \times \frac{1}{100} \times \frac{2.5}{20/4}}{0.01}$$

$$= 500 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}.$$

12. **Ans (C)**

Sum of stoichiometric coefficient of gaseous

$$\text{product} = \frac{n}{3} + \frac{2n}{3} = n$$

13. **Ans (D)**

Rate of diffusion of NH_3 will be faster than HCl .

14. **Ans (B)**

Milli equivalent of $\text{Na}_2\text{C}_2\text{O}_4$ = Milli equivalent of KMnO_4 reacted.

Total milli equivalent of KMnO_4 – Excess milli equivalent of KMnO_4 reacted with $\text{H}_2\text{C}_2\text{O}_4$.

$$= 45 \times 0.02 \times 5 - 10 \times 0.1 \times 2 = 2.5$$

$$\therefore 1000 \times \frac{w}{134} \times 2 = 25$$

$$\therefore w = 0.1675 \text{ gm}$$

$$\therefore \% \text{ purity of } \text{Na}_2\text{C}_2\text{O}_4 \text{ in sample}$$

$$= \frac{0.1675}{0.2} \times 100 = 83.75$$

PART-2 : CHEMISTRY

SECTION-II

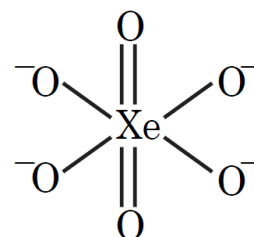
5. **Ans (3.00)**

gly^{-1} , bcac^{-1} , pn

6. **Ans (4.00)**

Fe , Sn , Zn , Pb

7. **Ans (1.33)**



$$\text{Bond order} = 8/6 = 1.33$$

8. **Ans (20.00)**

$$T_B = \frac{a}{Rb} = \frac{2.5}{0.08 \times 0.0625} = 500 \text{ K}$$

$$PV = nRT$$

$$2 \times V = 1 \times 0.08 \times 500$$

$$\Rightarrow V = 20 \text{ L}$$

9. **Ans (62.50)**

Coagulation value

$$= \frac{\text{No. of millimoles of electrolyte required}}{\text{Volume of colloidal solution in L}}$$

$$= \frac{20 \times 0.25}{80 \times 10^{-3}}$$

$$= 62.50$$

10. **Ans (7.50)**

Exp (1) : $r = K[B]^y$ as $[A]_0 > [B]_0$

$$\therefore r_{7/8} = 3 \times t_{1/2} \Rightarrow y = 1$$

Exp (2) : $r = K''[A]^x$ as $[A]_0 \ll [B]_0$

$$t_{7/8} = 7 \times t_{1/2} \Rightarrow x = 2$$

$$t_{1/2} = \frac{1}{K''[A]_0} = 10 K'' = K[B]_0$$

$$\therefore K'' = \frac{1}{10 \times 3}$$

From Exp (2) and (3)

$$t_{1/2} = \frac{1}{K''[A]_0} = \frac{1}{K[B]_0[A]_0}$$

$$\therefore a = \frac{10}{2 \times 2} = 2.5 \text{ and } b = 7 \times 2.5 = 17.5$$

PART-3 : MATHEMATICS

SECTION-I

1. **Ans (C)**

$$z_1^3 - 3z_1z_2^2 + 3iz_2z_1^2 - iz_2^3 = 2 + 11i$$

$$(z_1 + iz_2)^3 = 2 + 11i$$

$$\text{Similarly } (z_1 - iz_2)^3 = 2 - 11i$$

$$\text{Multiply } (z_1^2 + z_2^2)^3 = 4 + 12i$$

$$(z_1^2 + z_2^2)^3 = 125$$

$$z_1^2 + z_2^2 = 5$$

2. **Ans (C)**

$$x^3 + 27\cot^3x + 9x^2\cot x + 27x \cot^2x = \pi^3$$

$$(x + 3\cot x)^3 = \pi^3$$

$$\Rightarrow x + 3 \cot x = \pi$$

$$\Rightarrow \cot x = \frac{\pi}{3} - \frac{x}{3} \text{ (Now graph)}$$

3. **Ans (D)**

Required probability

$$= \frac{\frac{1}{5} \times \frac{1}{5}}{\frac{1}{5} \left(\frac{1}{6}\right) + \frac{1}{5} \left(\frac{1}{5}\right) + \frac{1}{5} \left(\frac{1}{5}\right) + \frac{1}{5} \times \frac{1}{5} + \frac{1}{5} \times \frac{1}{5}}$$

$$= \frac{6}{29}$$

4. **Ans (C)**

Line joining intersection point of tangents and middle point of the line joining both the points is parallel to axis of the parabola.

$$\text{Midpoint } (2, 4), \text{ slope of axis } \frac{4-1}{2+1} = 1,$$

$$\text{slope of directrix} = -1$$

5. **Ans (D)**

$$R = \{(a, b) : a^4 - b^4 \geq 0\}$$

$$\text{Let } (a, a) \in R$$

It is satisfying the relation condition

It will be reflexive

$$\text{Let } (a, b) \in R \Rightarrow a^4 - b^4 \geq 0$$

$$b^4 - a^4 \leq 0 \therefore (b, a) \notin R$$

It does not satisfy the symmetric relation condition.

$$\text{Let } (a, b) \in R \text{ and } (b, c) \in R$$

$$\therefore a^4 - b^4 \geq 0 \text{ and } b^4 - c^4 \geq 0 \Rightarrow (a^4 - c^4) \geq 0$$

$$\therefore (a, c) \in R$$

It will be transitive relation.

It is reflexive and transitive but not symmetric.

6. **Ans (B)**

$$\{(\bar{a} \times \bar{b}) \times \bar{a}\} \times \bar{b} + \bar{a} \times \{\bar{b} \times (\bar{a} \times \bar{b})\}$$

$$= [(\bar{a} \cdot \bar{a}) \bar{b} - (\bar{a} \cdot \bar{b}) \bar{a}] \times \bar{b} + \bar{a} \times [(\bar{b} \cdot \bar{b}) \bar{a} - (\bar{a} \cdot \bar{b}) \bar{b}]$$

$$= (-2(\bar{a} \cdot \bar{b})) \bar{a} \times \bar{b}$$

$$= (2(\bar{a} \cdot \bar{b})) \bar{b} \times \bar{a}$$

7. **Ans (A)**

$$\frac{1}{2} \left(\int_{-10}^{-1/3} (+1) dx + \int_{-1/3}^0 (-1) dx \right)$$

$$\frac{1}{2} \left([x]_{-10}^{-1/3} - [x]_{-1/3}^0 \right)$$

$$= \frac{14}{3}$$

8. **Ans (B)**

Let p is the statement that a positive number is prime and q is the statement that number has no divisor other than 1 and itself.

Contrapositive of $P \rightarrow q$ is $\neg q \rightarrow \neg P$

So contrapositive of the above statement is if a positive integer has divisors other than 1 and itself then it is not prime.

9. **Ans (B)**

$$\frac{1}{2}(2xdx + 2ydy) + \frac{\frac{xdy - ydx}{x^2}}{1 + \frac{y^2}{x^2}} = 0$$

$$\frac{1}{2}D(x^2 + y^2) + D \tan^{-1}\left(\frac{y}{x}\right) = 0$$

$$\text{Hence, } \frac{x^2 + y^2}{2} + \tan^{-1}\frac{y}{x} = C$$

$$\frac{c - x^2 - y^2}{2} = \tan^{-1}\frac{y}{x}$$

$$y = x \tan\left(\frac{c - x^2 - y^2}{2}\right)$$

10. **Ans (C)**

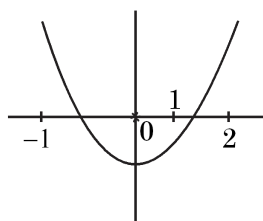
$$\begin{aligned} \because (\tan A - \tan B)^2 + (\tan B - \tan C)^2 + (\tan C - \tan A)^2 &> 0 \\ 2(\tan^2 A + \tan^2 B + \tan^2 C) - 2(\tan A \tan B + \tan B \tan C + \tan C \tan A) &> 0 \end{aligned}$$

$$3K - (\tan A \tan B \tan C)^2 > 0$$

$$K > 48$$

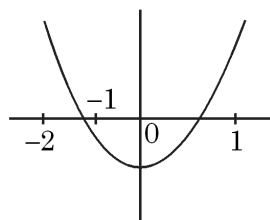
11. **Ans (A)**

Case (I)



$$\begin{aligned} f(-1) > 0, \quad f(1) < 0, \quad f(2) > 0 \\ 1 - a - 1 > 0, \quad 1 + a - 1 < 0, \quad 4 + 2a - 1 > 0 \\ a < 0, \quad a < 0, \quad 2a > -3 \\ -\frac{3}{2} < a < 0 \text{ since 'a' integer so } a = -1 \end{aligned}$$

Case (II)



$$\begin{aligned} f(-2) > 0, \quad f(-1) < 0, \quad f(1) > 0 \\ 4 - 2a - 1 > 0, \quad 1 - a - 1 < 0, \quad 1 + a - 1 > 0 \\ 2a < 3, \quad a > 0, \quad a > 0 \\ a < \frac{3}{2} \\ 0 < a < \frac{3}{2}, \text{ since 'a' integer so } a = 1 \end{aligned}$$

12. **Ans (C)**

$$\sin^{-1}\left(\frac{x^2 + 1}{2x}\right) \text{ is defined for}$$

$$-1 \leq \frac{x^2 + 1}{2x} \leq 1 \Rightarrow x = 1, -1$$

But square root is there so $x = 1$ is the only point.

$$\text{Now } \log_{[x]+1} \frac{|x|}{x} \text{ is defined for } x \neq 0 \text{ and}$$

$$[x] + 1 > 1$$

$$[x] > 0$$

$$x \geq 1$$

So common domain is $x = 1$

13. **Ans (C)**

$$2ae = 4 \Rightarrow ae = 2 \Rightarrow a^2 - b^2 = a^2 e^2 = 4$$

$$\text{Also, } p_1 p_2 = b^2 \Rightarrow b^2 = \left(\frac{5}{\sqrt{5}}\right) \left(\frac{1}{\sqrt{5}}\right) \Rightarrow b^2 = 1$$

$$\therefore a^2 = 5$$

$$\therefore a^2 + b^2 = 6$$

14. **Ans (B)**

$$b = a + (n-1)d, c = \frac{n}{2}(a+b)$$

$$\therefore d = \frac{(a-d)(a+b)}{a+b-x\left(\frac{a+b}{2}\right)} \Rightarrow d = \frac{2(a-b)}{2-nx}$$

$$\Rightarrow 2d - ndx = -2(n-1)d$$

$$\Rightarrow x = 2$$

15. **Ans (D)**

$$T_{r+1} = {}^{25}C_r (ax^{1/2})^{25-r} (bx^{-1/3})^r$$

$$= {}^{25}C_r a^{25-r} b^r x^{\frac{25-r}{2} - \frac{r}{3}}$$

$$\text{or } \frac{25-r}{2} - \frac{r}{3} = 0 \Rightarrow 75 - 3r = 2r \Rightarrow r = 15$$

$$T_{r+1} = {}^{25}C_{15} (a)^{10} (b)^{15}$$

Now, $a > 0, b > 0 \therefore$ by AM $\geq$ GM

$$\frac{10a + 15b}{25} \geq (a^{10} b^{15})^{1/25}$$

$$\text{or } a^{10} b^{15} \leq \left(\frac{1}{25}\right)^{25}$$

$$\therefore T_{r+1(\max)} = {}^{25}C_{10} \left(\frac{1}{25}\right)^{25}$$

16. **Ans (D)**

The series can be re-written as

$$\tan^{-1}(2) + \tan^{-1}\left(\frac{4}{13}\right) + \tan^{-1}\left(\frac{6}{73}\right) + \tan^{-1}\left(\frac{8}{241}\right) \dots \infty$$

Hence the general term T_r

$$= \tan^{-1}\left(\frac{2r}{1-r^2+r^4}\right)$$

$$S_{\infty} = \lim_{n \rightarrow \infty} \sum_{r=1}^n \tan^{-1}\left(\frac{2r}{1-r^2+r^4}\right)$$

$$S_{\infty} = \lim_{n \rightarrow \infty} \sum_{r=1}^n \tan^{-1}\left(\frac{r(r+1)-r(r-1)}{1+(r(r+1))(r(r-1)))}\right)$$

$$S_{\infty} = \lim_{n \rightarrow \infty} \sum_{r=1}^n (\tan^{-1}(r(r+1)) - \tan^{-1}(r(r-1)))$$

$$S_{\infty} = \lim_{n \rightarrow \infty} (\tan^{-1}(n(n+1)) - \tan^{-1}(0))$$

$$S_{\infty} = \frac{\pi}{2} - 0 = \frac{\pi}{2}$$

17. **Ans (A)**

Since point $(-1, 1)$ lies on $3x - 2y + 5 = 0$

$\Rightarrow$ locus of point is line perpendicular to

$3x - 2y + 5 = 0$ and passing through $(-1, 1)$

18. **Ans (C)**

$$g'(x) = 2f'\left(\frac{x^4}{2}\right) \cdot \frac{4x^3}{2} + f'(6-x^4)(-4x^3)$$

$$= 4x^3 \left(f'\left(\frac{x^4}{2}\right) - f'(6-x^4) \right)$$

$g'(x)$ should be negative for decreasing nature.

Two cases are possible

$$(i) x^3 < 0 \text{ and } f'\left(\frac{x^4}{2}\right) - f'(6-x^4) > 0$$

$$\Rightarrow x < 0 \quad f'\left(\frac{x^4}{2}\right) > f'(6-x^4)$$

But $f''(x)$ so $f'(x)$ is decreasing

$$\frac{x^4}{2} < 6 - x^4$$

$$x^4 \left(1 + \frac{1}{2}\right) < 6$$

$$x^4 \left(\frac{3}{2}\right) < 6$$

$$x^4 < 4 \Rightarrow (x^2+2)(x^2-2) < 0$$

$$-\sqrt{2} < x < \sqrt{2} \quad x \in (-\sqrt{2}, \sqrt{2})$$

$$(ii) x^3 > 0 \quad f'\left(\frac{x^4}{2}\right) - f'(6-x^4) < 0$$

$$\Rightarrow x > 0 \quad \Rightarrow x \in (-\infty, -\sqrt{2}) \cup (\sqrt{2}, \infty)$$

So $x \in (-\sqrt{2}, \sqrt{2})$

19. **Ans (D)**

$$\text{adj}(4\text{adj}A^2) = 16\text{adj}(\text{adj}A^2)$$

$$\text{We know } \text{adj}(\text{adj}A) = |A|^{(n-2)}(A)$$

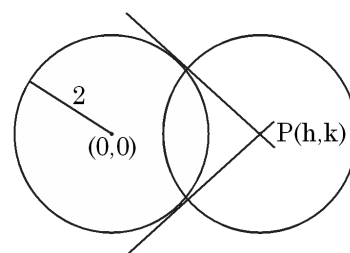
$$\text{adj}(\text{adj}|A|A^2) = 16 \times 4 \times A^2 = 64A^2$$

20. **Ans (A)**

Equation of chord of contact w.r.t. P

$$xh + yk = 4 \quad \dots(1)$$

Equation of common chord



$$(\lambda - 2)x + (3\lambda + 1)y = -9 \quad \dots(2)$$

Compare (i) and (ii)

$$\frac{h}{\lambda - 2} = \frac{k}{(3\lambda + 1)} = \frac{4}{-9}$$

$$\Rightarrow 27h - 9k = 28$$

$$\therefore 27x - 9y = 28$$

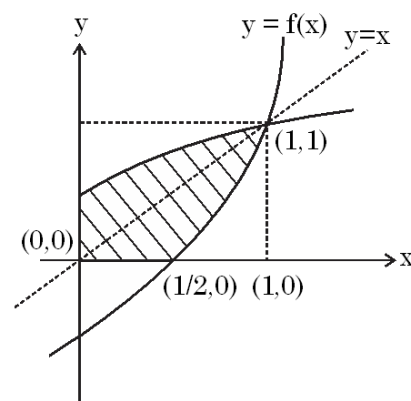
PART-3 : MATHEMATICS

SECTION-II

1. **Ans (9.50)**

$$f(x) = \frac{1}{4}(4x^3 + x - 1)$$

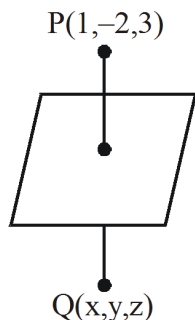
$$= \left(x - \frac{1}{2}\right) \left(x^2 + \frac{x}{2} + \frac{1}{2}\right)$$



$$A = 2 \left[\frac{1}{2} \times 1 \times 1 - \int_{1/2}^1 f(x) dx \right]$$

$$= 1 - \frac{1}{2} \int_{1/2}^1 (4x^3 + x - 1) dx = \frac{19}{32}$$

2. **Ans (23.33)**



PQ is parallel to normal vector

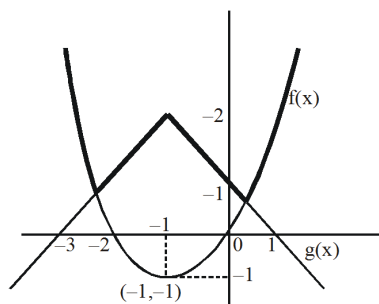
$$\frac{x-1}{1} = \frac{y+2}{-1} = \frac{z-3}{1} = \lambda$$

$$\Rightarrow x = \lambda + 1, y = -2 - \lambda, z = 3 + \lambda$$

$$\text{Mid point lies on plane} \Rightarrow \lambda = \frac{2}{3}$$

$$Q\left(\frac{5}{3}, -\frac{8}{3}, \frac{11}{3}\right) \Rightarrow OQ = \sqrt{\frac{70}{3}}$$

3. **Ans (5.00)**



$$p = 3, q = 2$$

$$p + q = 5$$

4. **Ans (3.99)**

$$\frac{\sum x_i}{20} = 10 \Rightarrow \sum x_i = 200$$

$$\frac{\sum x_i^2}{20} - 100 = 4$$

$$\sum x_i^2 = 104 \times 20 = 2080$$

$$\text{Actual mean} = \frac{200 - 9 + 11}{20} = \frac{202}{20}$$

$$\text{Variance} = \frac{2080 - 81 + 121}{20} - \left(\frac{202}{20}\right)^2$$

$$= \frac{2120}{20} - (10.1)^2$$

$$= 106 - 102.01$$

$$= 3.99$$

5. **Ans (1.50)**

For non trivial solution

$$\Delta = \begin{vmatrix} a+2t & b & c \\ b & c+2t & a \\ c & a & b+2t \end{vmatrix} = 0$$

Clearly Δ is cubic in t and has 3 roots $K = 3$

6. **Ans (8.50)**

$$I = \int \frac{\frac{e^{x/2}}{2} \sin 3x + 3e^{x/2} \cos 3x}{5 + e^{x/2} \sin 3x} dx$$

Take $5 + e^{x/2} \sin 3x = t$

$$\left(\frac{e^{x/2}}{2} \sin 3x + 3e^{x/2} \cos 3x\right) dx = dt$$

$$I = \int \frac{dt}{t} = \ln |5 + e^{x/2} \sin 3x| + C$$

$$\text{Hence } \lambda = 5, \mu = \frac{1}{2} \text{ and } \gamma = 3$$

7. **Ans (3.75)**

$$x_1 + x_2 + x_3 + x_4 = 15, \text{ No. of ways} = {}^{18}C_3$$

$$x_1 + x_2 + x_3 + x_4 = 10, \text{ No. of ways} = {}^{13}C_3$$

$$x_1 + x_2 + x_3 + x_4 = 12, \text{ No. of ways} = {}^{15}C_3$$

$$\text{Total No. of ways} = {}^{18}C_3 + {}^{13}C_3 + {}^{15}C_3 \Rightarrow K = 3$$

8. **Ans (0.00)**

For homogeneous equation $\frac{d^2y}{dx^2} = 0$ at point lying on the curve.

9. **Ans (4.00)**

Let $P(\alpha, \beta)$ then

$$\alpha - \beta + 2 = 0$$

$$\text{Equation of C.O.C for } \frac{x^2}{6} - \frac{y^2}{2} = 1$$

$$\text{is } \frac{\alpha - x}{6} - \frac{\beta y}{2} = 1$$

$$\text{or } \alpha x - 3\beta y = 6$$

$$\text{or } \alpha x - 3(\alpha + 2)y - 6 = 0$$

$$\text{or } \alpha(x - 3y) - 6y - 6 = 0$$

$$\therefore x = -3, y = -1$$

$$|x + y| = 4$$

10. **Ans (4.00)**

$$\lambda = \lim_{x \rightarrow 0} \frac{x \left(1 - \frac{11}{2} \cos x\right) + \frac{9}{2} \sin x}{\sin^3 x}$$

Using series $\lambda = 2$

$$\lim_{x \rightarrow 2} \frac{\sqrt{x^2 - 4}}{\sqrt{x - 2} + \sqrt{x} - \sqrt{2}} = 2$$